ANOVA

One-way ANOVA

$$SS_{Total} = \sum_i \sum_j \left(Y_{ij} - \bar{Y}_{..}\right)^2$$
$$Total\ df = N - 1$$
$$SS_{treatment} = \sum_i r(\bar{Y}_{i.} - \bar{Y}_{..})^2$$
$$Treatment\ df = a - 1$$
$$SSE = \sum_i \sum_j \left(Y_{ij} - \bar{Y}_{i.}\right)^2$$
$$Error\ df = N - a$$

Two-way ANOVA without interaction

$$SS_{Total} = \sum_i \sum_j \left(Y_{ij} - \bar{Y}_{..}\right)^2$$
$$Total\ df = N - 1$$
$$SS_{treatment} = \sum_i b(\bar{Y}_{i.} - \bar{Y}_{..})^2$$
$$Treatment\ df = a - 1$$
$$SS_{block} = \sum_j a\left(\bar{Y}_{.j} - \bar{Y}_{..}\right)^2$$
$$Blocking\ df = b - 1$$
$$SSE = \sum_i \sum_j \left(Y_{ij} - \bar{Y}_{i.} - \bar{Y}_{.j} - \bar{Y}_{..}\right)^2$$
$$Error\ df = (a - 1)(b - 1)$$

One-way ANOVA with interaction

$$SS_{Total} = \sum_i \sum_j \sum_k \left(Y_{ijk} - \bar{Y}_{...}\right)^2$$
$$Total\ df = N - 1$$
$$SS_A = \sum_i rb(\bar{Y}_{i..} - \bar{Y}_{...})^2$$
$$A\ df = a - 1$$
$$SS_B = \sum_j ra\left(\bar{Y}_{.j.} - \bar{Y}_{...}\right)^2$$
$$B\ df = b - 1$$
$$SS_{AB} = \sum_i \sum_j r\left(\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...}\right)^2$$
$$AB\ df = (a - 1)(b - 1)$$
$$SSE = \sum_i \sum_j \sum_k \left(Y_{ijk} - \bar{Y}_{ij.}\right)^2$$
$$Error\ df = abr - 1$$

NON-PARAMETRIC

Friedman Statistic:

$$F_r = \left[\frac{12}{bk(k+1)} \sum_{j=1}^{k} T_j^2\right] - 3b(k+1), \mathrm{df} = k - 1$$

Kruskal Wallis Statistic:

$$H = \left[\frac{12}{n_T(n_T+1)} \sum_{i=1}^{k} \frac{T_i^2}{n_i}\right] - 3(n_T + 1), \mathrm{df} = k - 1$$

Mann-Whitney-Wilcoxon Test Statistic (normal approximation, n_1 and/OR $n_2 \geq 10$):

$$z = \frac{T-\mu_T}{\sigma_T} \text{ where } \mu_T = \frac{n_1(n_1+n_2+1)}{2}, \sigma_T = \sqrt{\frac{n_1 n_2(n_1+n_2+1)}{12}}$$

Spearman Rank correlation ($n < 10$):

$$r_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2 - 1)}$$

Spearman Rank correlation (normal approximation, $n \geq 10$):

$$z = \frac{r_s - 0}{\sqrt{\frac{1}{n-1}}} = r_s\sqrt{n-1}$$

Wilcoxon Signed Rank Sum Statistic (normal approximation, $n \geq 10$):

$$z = \frac{T-\mu_T}{\sigma_T} \text{ where } \mu_T = 0, \sigma_T = \sqrt{\frac{n(n+1)(2n+1)}{6}}$$

SIMPLE LINEAR REGRESSION

Sum of squares:

$$SSR = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

Measures of association:

$$r_{xy} = \frac{SS_{xy}}{\sqrt{SS_x SS_y}}$$

Standard error of the regression slope:

$$s_{\beta_1} = \frac{s_\varepsilon}{\sqrt{SS_x}}$$

$(1-\alpha)$% confidence interval for β_j:

$$\widehat{\beta_J} \pm t_{\frac{\alpha}{2}, n-p-1} s_{\beta_j}$$

$(1-\alpha)$% confidence interval for $\overline{Y}|x_g$:

$$\hat{y} \pm t_{\frac{\alpha}{2}, n-p-1} s_\varepsilon \sqrt{\frac{1}{n} + \frac{\left(x_g - \bar{x}\right)^2}{\sum (x_i - \bar{x})^2}}$$

$(1-\alpha)$% prediction interval for $Y|x_g$:

$$\hat{y} \pm t_{\frac{\alpha}{2}, n-p-1} s_\varepsilon \sqrt{1 + \frac{1}{n} + \frac{\left(x_g - \bar{x}\right)^2}{\sum (x_i - \bar{x})^2}}$$

MULTIPLE REGRESSION

Multiple regression model (for the sake of defining n and p):

$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i,1} + \hat{\beta}_2 x_{i,2} + \cdots + \hat{\beta}_p x_{i,p}$ for $i = 1, 2, 3, \ldots, n$

Standard error of estimate:

$$s_\varepsilon = \sqrt{MSE} = \sqrt{\frac{SSE}{n-p-1}}$$

Multiple coefficient of determination:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

Adjusted multiple coefficient of determination:

$$R^2_{adj} = 1 - \frac{n-1}{n-p-1} \times \frac{SSE}{SST} = 1 - \frac{(n-1)}{(n-p-1)}\ (1-R^2)$$

Coefficient of partial determination

$$R^2_{partial} = \frac{SSE_{reduced} - SSE_{full}}{SSE_{reduced}}$$

Partial *F*-statistic

$$F_{partial} = \frac{\dfrac{\left(SSE_{reduced} - SSE_{full}\right)}{r}}{SSE_{full}\big/(n-p-1)}$$

LOGISTIC REGRESSION

logit: $log\left(\frac{p}{1-p}\right)$

Inverse logit: $p = \dfrac{1}{1+exp^{-(\beta_0+\beta_1 x+\cdots)}}$

TIME SERIES FORMULAE

AUTOCORRELATION

ACF at lag h: $\rho(h) = \dfrac{\sum_{t=h+1}^{T}(y_t-\bar{y})(y_{t-h}-\bar{y})}{\sum_{t=1}^{T}(y_t-\bar{y})^2}$

Ljung-Box Test Statistic: $Q_{LB}(m) = n(n+2) \times \sum_{h=1}^{m} \frac{\hat{\rho}(h)^2}{n-h} \; h = 1, 2, \dots, m$

SIMPLE FORECASTING METHODS

Average method: $\hat{y}_{t+h|T} = \bar{y} = (y_1 + y_2 + \cdots + y_T)/T$

Naïve method: $\hat{y}_{t+h|T} = y_T$

Seasonal naïve method: $\hat{y}_{t+h|T} = y_{T+h-m(k+1)}$ where k is the integer part of (h-1)/m

Drift method: $\hat{y}_{t+h|T} = y_T + \frac{h}{T-1} \sum_{t=2}^{T} (y_t - y_{t-1}) = y_T + h(\frac{y_T - y_1}{T-1})$

MEASURES OF FORECAST ACCURACY

Mean absolute error: $MAE = \sum_{t=1}^{n} \frac{|Y_t - \hat{Y}_t|}{n}$

Mean squared error: $MSE = \sum_{t=1}^{n} \frac{(Y_t - \hat{Y}_t)^2}{n}$

95% Prediction interval: $\hat{y}_{t+h|T} \pm 1.96\hat{\sigma}_h$

80% Prediction interval: $\hat{y}_{t+h|T} \pm 1.28\hat{\sigma}_h$

General Prediction interval: $\hat{y}_{t+h|T} \pm c\hat{\sigma}_h$ where c is read from table

TRANSFORMATIONS

Box-Cox transformations: $w_t = \begin{cases} \log(y_t) & if\ \lambda = 0 \\ (y_t^{\lambda} - 1)/\lambda & otherwise \end{cases}$

Differencing:

First-order at lag 1: $w_t = y_t - y_{t-1} \quad t = 2, 3, \dots, T$

Seasonal at lag p: $w_t = y_t - y_{t-p} \quad t = p+1, p+2, \dots, T$

CLASSICAL DECOMPOSITION MODEL

Multiplicative model formulation: $Y_t = T_t \times S_t \times R_t$ assuming $C_t = 1$ i.e. it is negligible

Trend Equation:
$T_t = \hat{\beta}_0 + \hat{\beta}_1 t$

Forecast equation:
$\hat{Y}_t = \left(\hat{\beta}_0 + \hat{\beta}_1 t\right) \times SI_t$

EXPONENTIAL SMOOTHING MODELS

Simple Exponential smoothing

Forecast equation: $\hat{y}_{t+h|t} = l_t$ **Level equation:** $l_t = \alpha y_t + (1-\alpha) l_{t-1}$

Level smoothing parameter $0 \leq \alpha \leq 1$

Holt's linear trend

Forecast equation: $\hat{y}_{t+h|t} = l_t + h b_t$

Level equation: $l_t = \alpha y_t + (1-\alpha)(l_{t-1} + b_{t-1})$

Trend equation: $b_t = \beta^*(l_t - l_{t-1}) + (1-\beta^*) b_{t-1}$

Level smoothing parameter: $0 \leq \alpha \leq 1$ **Trend smoothing parameter:** $0 \leq \beta^* \leq 1$

Holt-Winters seasonal

Forecast equation: $\hat{y}_{t+h|t} = (l_t + h b_t) s_{t+h-m(k+1)}$

Level equation: $l_t = \alpha \frac{y_t}{s_{t-m}} + (1-\alpha)(l_{t-1} + b_{t-1})$

Trend equation: $b_t = \beta^*(l_t - l_{t-1}) + (1-\beta^*) b_{t-1}$

Seasonal equation: $s_t = \gamma \frac{y_t}{(l_{t-1} + b_{t-1})} + (1-\gamma) s_{t-m}$

Level smoothing parameter: $0 \leq \alpha \leq 1$ **Trend smoothing parameter:** $0 \leq \beta^* \leq 1$

Seasonal smoothing parameter: $0 \leq \gamma \leq 1$

ARIMA MODELS

Stationary AR(p) model:

Stationary AR(1) model:

$E[X_t] = \mu = \frac{\phi_0}{1-\phi_1- \phi_2-\cdots-\phi_p}$

$\rho(1) = \frac{\phi_1 V[X_t]}{V[X_t]} = \phi_1$

$E[X_t] = \mu = \frac{\phi_0}{1-\phi_1}$

$\rho(h) = \phi_1^h$ $\quad h$ = 1, 2,…,n

$V[X_t] = \frac{\sigma^2}{1-\phi_1^2}$

<u>MA(q) model</u>:

$E[X_t] = \mu$

$V[X_t] = \sigma^2(1 + \theta_1^2 + \theta_2^2 + \cdots + \theta_q^2)$

<u>MA(1) model</u>:

$E[X_t] = \mu$

$V[X_t] = \sigma^2(1 + \theta_1^2)$

$\rho(1) = \frac{\theta_1}{1+\theta_1^2}$

<u>Stationary ARMA(p,q) model</u>: $\mu = \frac{\phi_0}{1-\phi_1-\phi_2-\cdots-\phi_p}$

<u>Akaike Information Criterion (AIC)</u>:

$AIC(k) = -2\log(L) + 2(p + q + k + 1)$